

Cuaderno de Prácticas

Álgebra

Grado en Ingeniería Informática

CURSO 2025/26

Datos personales

Nombre y apellidos : Francisco Javier Martín - Lunas Escobar

DNI : 26268082

Grupo de teoría : A

Grupo de prácticas :1

Práctica 1

In[78]:=

```
Euclides2[m_]:=Module[{p,q,d},If[Length[m]==3,p[x_]:=PolynomialMod[m[[1]],m[[3]]];  
q[x_]:=PolynomialMod[m[[2]],m[[3]]];  
d[x_]:=PolynomialQuotientRemainder[p[x],q[x],x,Modulus->m[[3]]];p[x_]:=m[[1]];q[x_]:=m[[2]];  
d[x_]:=PolynomialQuotientRemainder[p[x],q[x],x];];Print[p[x]," = (" ,q[x]," ) · (" ,d[x] [[1]],"  
+ ",d[x] [[2]]];If[Exponent[d[x] [[2]],x]<0,Print["El máximo común divisor es"]];  
Return[q[x]],If[Length[m]==3,Euclides2[{q[x],d[x] [[2]],m[[3]]}],Euclides2[{q[x],d[x] [[2]]}]]];
```

Ejercicio 1. A) Ordinaria 2 del curso 24/25

A) Aplicar si es posible, el algoritmo de Euclides para calcular en $\mathbb{Z}_7[X]$ divisor de los polinomios

$$p(x)=32+80x+96x^2+64x^3+24x^4+4x^5$$

$$q(x)=-2x^3+2x^2+8x+12$$

```
In[ ]:= p[x_]:=PolynomialMod[32+80x+96x^2+64x^3+24x^4+4x^5,7]
q[x_]:=PolynomialMod[-2x^3+2x^2+8x+12,7]
p[x]
q[x]
```

```
Out[ ]:=
4 + 3 x + 5 x^2 + x^3 + 3 x^4 + 4 x^5
```

```
Out[ ]:=
5 + x + 2 x^2 + 5 x^3
```

```
In[ ]:= c1[x_]:=PolynomialMod[PolynomialQuotient[p[x],q[x],x],7]
r1[x_]:=PolynomialMod[PolynomialRemainder[p[x],q[x],x],7]
c1[x]
r1[x]
```

```
Out[ ]:=
2 + 5 x^2
```

```
Out[ ]:=
1 + x + 4 x^2
```

```
In[ ]:= c2[x_]:=PolynomialMod[PolynomialQuotient[q[x],r1[x],x],7]
r2[x_]:=PolynomialMod[PolynomialRemainder[q[x],r1[x],x],7]
c2[x]
r2[x]
```

```
Out[ ]:=
5 + 3 x
```

```
Out[ ]:=
0
```

```
In[ ]:= Euclides2[{p[x],q[x],7}]
```

$$4 + 3 x + 5 x^2 + x^3 + 3 x^4 + 4 x^5 = (5 + x + 2 x^2 + 5 x^3) \cdot (2 + 5 x^2) + 1 + x + 4 x^2$$

$$5 + x + 2 x^2 + 5 x^3 = (1 + x + 4 x^2) \cdot (5 + 3 x) + 0$$

El máximo común divisor es

```
Out[ ]:=
1 + x + 4 x^2
```

El máximo común divisor es el último resto no nulo

$1 + x + 4 x^2$ único salvo asociados

B) Razonar la veracidad o falsedad de la siguiente afirmación: “En el anillo $\mathbb{Z}_7[x]$ de polinomio $2x^3 - 2x^2 - 8x - 12$ es irreducible”.

```
In[ ]:= r[x_] := PolynomialMod[2x^3 - 2x^2 - 8x - 12, 7]
r[x]
```

```
Out[ ]:= 2 + 6 x + 5 x^2 + 2 x^3
```

```
In[ ]:= Factor[r[x]]
IrreduciblePolynomialQ[r[x]]
```

```
Out[ ]:= (1 + 2 x) (2 + 2 x + x^2)
```

```
Out[ ]:= False
```

Falso, no es irreducible

Ejercicio 1. A) Extraordinaria 2 del curso 24/25

A) Factorizar en irreducibles y calcular las raíces del polinomio $p(x) = 12 - 60x + 16x^2 - 80x^3 + 4x^4 - 20x^5$ en $\mathbb{Z}[x]$, $\mathbb{Z}_{11}[x]$, $\mathbb{Q}[x]$ y $\mathbb{C}[x]$, sabiendo que $p(x)$ no tiene raíces enteras

```
In[63]:= Factor[12 - 60x + 16x^2 - 80x^3 + 4x^4 - 20x^5]
```

```
Out[63]:= -4 (-1 + 5 x) (1 + x^2) (3 + x^2)
```

El 4 debe darse descompuesto como 2^2 al tratarse de los números enteros

$\mathbb{Q}[x]$

```
In[66]:= "La única raíz, simple de hecho, es 1/5"
-4*5 (-1/5 + x) (1+x^2) (3+x^2)
```

```
Out[66]:= La única raíz, simple de hecho, es 1/5
```

```
Out[67]:= -20 (-1/5 + x) (1 + x^2) (3 + x^2)
```

Z11[x]

In[71]:= **Factor**[-20x⁵+4x⁴-80x³+16x²-60x+12,Modulus→11]

Out[71]=
 $2 (9 + x) (6 + 6 x + 6 x^3 + x^4)$

C[x]

In[77]:= **Roots**[-20x⁵+4x⁴-80x³+16x²-60x+12==0,x,Complex]

Roots: Options expected (instead of Complex) beyond position 2 in

Roots[12 - 60 x + 16 x² - 80 x³ + 4 x⁴ - 20 x⁵ == 0, x, Complex]. An option must be a rule or a list of rules. [i](#)

Out[77]=
 $x = -\frac{1}{5} \mid x = -i \mid x = i \mid x = -i\sqrt{3} \mid x = i\sqrt{3}$

$-20(x-1/5)(x+i)(x-i)(x+isqr(3))(x-isqr(3))$

Ejercicio 13.7 Dados los siguientes polinomios

In[82]:= **p13**[x] := 2x⁴+5x³+6x²+4x+1
q13[x] := 4x⁶+8x⁵+5x⁴-x³-4x²-x+1

1)

In[84]:= **Euclides2**[{**p13**[x],**q13**[x],3}]

$$1 + x + 2x^3 + 2x^4 = (1 + 2x + 2x^2 + 2x^3 + 2x^4 + 2x^5 + x^6) \cdot (0) + 1 + x + 2x^3 + 2x^4$$

$$1 + 2x + 2x^2 + 2x^3 + 2x^4 + 2x^5 + x^6 = (1 + x + 2x^3 + 2x^4) \cdot (2 + 2x + 2x^2) + 2 + x + x^2 + 2x^3$$

$$1 + x + 2x^3 + 2x^4 = (2 + x + x^2 + 2x^3) \cdot (2 + x) + 0$$

El máximo común divisor es

Out[84]=
 $2 + x + x^2 + 2x^3$

2)

Enteros

```
In[85]:= Factor[p13[x]]
Factor[q13[x]]
PolynomialGCD[p13[x],q13[x],x]
PolynomialLCM[p13[x],q13[x],x]
```

Out[85]= $(1+x)(1+2x)(1+x+x^2)$

Out[86]= $(1+x)^2(-1+2x)^2(1+x+x^2)$

Out[87]= 1

Out[88]= $x(1+2x)(1-x-4x^2-x^3+5x^4+8x^5+4x^6)$

$\mathbb{Z}_7[x]$

```
In[89]:= Factor[p13[x],Modulus→7]
Factor[q13[x],Modulus→7]
```

Out[89]= $2(1+x)(3+x)(4+x)(5+x)$

Out[90]= $4(1+x)^2(3+x)^3(5+x)$

$\mathbb{R}[x]$

```
In[93]:= Factor[p13[x],Extension→Real]
Factor[q13[x],Extension→Real]
```

Out[93]= $\text{Factor}[1+4x+6x^2+5x^3+2x^4, \text{Extension} \rightarrow \text{Real}]$

Out[94]= $\text{Factor}[1-x-4x^2-x^3+5x^4+8x^5+4x^6, \text{Extension} \rightarrow \text{Real}]$

$\mathbb{Q}[x]$

```
In[97]:= Factor[p13[x],Extension→Rational]
Factor[q13[x],Extension→Extension]
```

Out[97]= $\text{Factor}[1+4x+6x^2+5x^3+2x^4, \text{Extension} \rightarrow \text{Rational}]$

Out[98]= $\text{Factor}[1-x-4x^2-x^3+5x^4+8x^5+4x^6, \text{Extension} \rightarrow \text{Extension}]$